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A PRIVACY-PRESERVING EDGE TINYML SYSTEM FOR ACOUSTIC COMFORT ESTIMATION USING HYBRID DSP-CNN FUSION AND VISUAL FEEDBACK

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Presentation Outline

1. Motivation and Problem
2. Objectives
3. Scope and Applications
4. Literature Review
5. System Methodology
6. TinyML Development
7. Results and Validation
8. Privacy Boundary
9. Future Enhancements
10. Conclusion

Motivation: Acoustic Comfort Needs Intelligence

Current problem

- Conversations and activities increase noise.
- Sound meters need people to keep checking them.
- Loud alarms can disturb the environment further.
- Direct warnings can make people feel uncomfortable.

Proposed response

- Monitor the sound level continuously.
- Identify speech and other noise activities.
- Give simple and calm visual feedback.
- Help people control their noise naturally.

SENSE $\longrightarrow$ **INTERPRET** $\longrightarrow$ **RESPOND**

Goal: self-regulating shared spaces through objective visual feedback.

Objectives

Main Objective

Design, develop, and validate a privacy-preserving edge acoustic-comfort indicator using hybrid digital signal processing and embedded deep learning.

Secondary Objectives

- Measure ambient sound levels and extract acoustic features in real time.
- Classify stationary background, speech-dominant, and non-speech activity using a TinyML CNN.
- Fuse deterministic DSP metrics with probabilistic model predictions.
- Provide adaptive visual feedback and anonymous dashboard analytics.

Scope and System Boundary

Included

- Capture sound at 16 kHz
- Measure sound level and key features
- Classify sounds using INT8 DS-CNN
- Combine results to determine noise state
- Show feedback on TFT, LEDs, and dashboard

Boundaries

- Raw audio is never stored or sent.
- The prototype is an acoustic-comfort indicator, not a certified legal sound-level meter.
- Dashboard needs a local Wi-Fi connection.
- Secure MQTT communication is future work.

Project Applications

01
Classrooms

02
Libraries

03
Study rooms

04
Laboratories

05
Shared offices

06
Collaborative zones

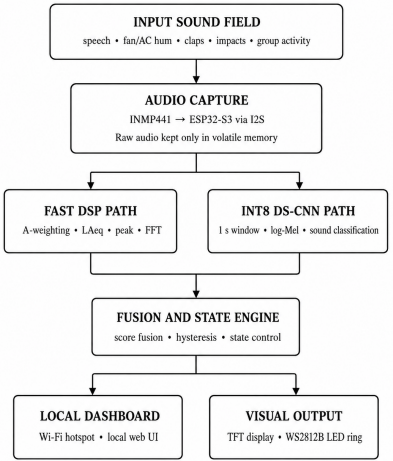
Literature Review and Research Gap

Prior work	Approach	Key finding	Gap addressed
IoT noise monitoring	dBA sensing and Wi-Fi dashboard	Live monitoring and logging	Adds sound-type awareness and local intelligence
TinyML anomaly detection	On-device inference	Low latency and low power	Adds stable and interpretable comfort states
Irrelevant-speech studies	Laboratory cognition experiments	Speech increases cognitive effort	Converts findings into practical feedback
Large-scale audio CNNs	YAMNet/AudioSet classification	Strong audio representations	Distills knowledge into an MCU-ready DS-CNN

Research Gap

Prior systems rarely combine calibrated DSP, TinyML classification, stable state logic, local privacy, and visual feedback in one low-cost embedded device.

Methodology - [1] Overall System Block Diagram



Methodology - [2] Core Hardware Components

ESP32-S3 N16R8

Core: dual-core 240 MHz Xtensa LX7.

Memory: 16 MB flash and 8 MB PSRAM.

Role: real-time DSP, INT8 inference, state control, display, and web service.

INMP441 I2S MEMS Microphone

Type: omnidirectional 24-bit digital microphone.

Interface: I2S digital audio.

Role: 16 kHz mono capture without an analog conversion stage.

Why this pair?

Digital I2S acquisition reduces analog noise, while the ESP32-S3 provides enough compute and memory for parallel DSP, TinyML, and local visualization.

Methodology - [3] Feedback and Power Hardware

1.8-inch TFT SPI Display

Interface: high-speed SPI.

Role: displays live disturbance state, dBA level, score, and system status.

WS2812B RGB LED Strip

Type: individually addressable LEDs.

Role: immediate ambient feedback through state-specific colors and animations.

Supporting hardware

- TP4056 charging/protection module
- 18650 Li-ion battery supply

Physical integration

- Compact 3D enclosure
- Separated sensor/display placement

Methodology - [4] Firmware and On-Device Software

Firmware and execution

- Arduino IDE and ESP32 toolchain
- FreeRTOS tasks and dual-core scheduling
- I2S DMA audio acquisition
- Ucglib TFT rendering and LED control

On-device processing

- ESP-DSP FFT and biquad filtering
- TensorFlow Lite for Microcontrollers
- Fully quantized INT8 inference
- Local HTTP/JSON status service

Scheduling principle

Fast DSP updates every 100 ms; one-second CNN inference does not block the feedback loop.

Methodology - [5] Offline AI and Dashboard Stack

Offline AI pipeline

- Python, TensorFlow, Keras, and Librosa
- Audio preprocessing and embeddings
- Teacher-student knowledge distillation
- DS-CNN training and INT8 quantization

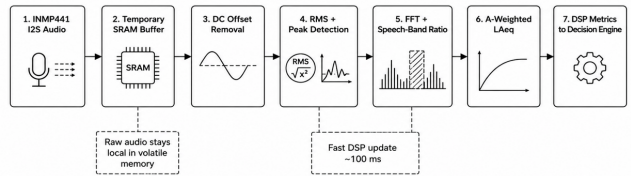
Visualization

- HTML5 Canvas and JavaScript
- Board-hosted responsive dashboard
- Asynchronous HTTP polling
- Anonymous scalar metrics only

Training occurs offline; operational inference occurs entirely on the edge device.

Methodology - [6] Audio Capture and DSP Pipeline

- 16 kHz mono audio capture using INMP441 and I2S DMA.
- Temporary PCM storage in an SRAM ring buffer.
- DC offset removal, windowing, RMS and peak calculation.
- FFT analysis for speech-band ratio and spectral flux.
- A-weighting and calibration for L_{Aeq} estimation.



Methodology - [7] Fast DSP Path

Update cadence

100 ms

1,600 samples per block at 16 kHz

Physical acoustic features

- A-weighted L_{Aeq}
- Peak, RMS, and crest factor
- 256-point FFT
- Speech-band energy ratio
- Spectral flux and spike evidence

Purpose: fast, deterministic reaction to loudness and transient events.

Methodology - [8] Log-Mel and TinyML Path

Inference window

1 s

16,000 samples, updated continuously

Feature tensor construction

- 512-point FFT with 320-sample hop
- 64 Mel filter-bank channels
- Log compression and normalization
- $49 \times 64 \times 1$ input tensor
- Three-class INT8 DS-CNN inference

Purpose: identify acoustic context that loudness alone cannot explain.

Methodology - [9] Teacher-Student Model Training

1. Transfer learning

- Reuse a model trained on a large audio dataset.
- Adapt learned representations to three project classes.

2. YAMNet teacher

- Frozen MobileNetV1-based audio classifier.
- Produces 1,024-D embeddings and soft targets.

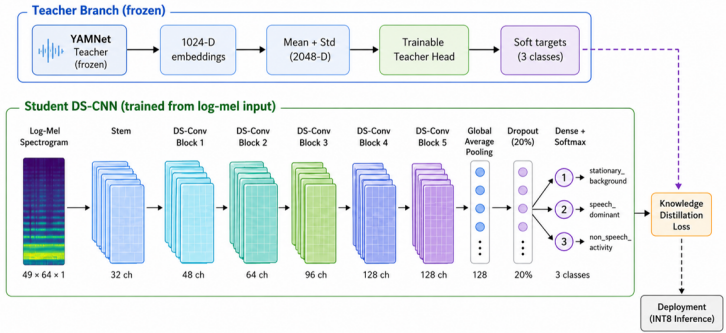
3. DS-CNN student

- Fast and lightweight; optimized for microcontrollers.
- Uses depthwise separable convolutions.

Knowledge Distillation

YAMNet is too large for microcontrollers. Knowledge Distillation trains a compact **Student** model to match the soft probability outputs of a heavy **Teacher** model. This compresses deep-learning intelligence so it can run directly on the ESP32-S3.

Methodology - [10] Teacher-Student DS-CNN Architecture



- Multiple student configurations were evaluated during training; the model with the lowest Knowledge Distillation loss was deployed on the ESP32-S3.

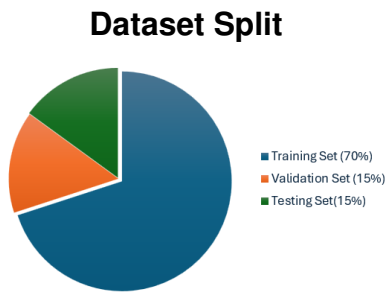
Methodology - [11] Dataset Overview

Dataset preparation

- Collected 12,466 environmental audio clips
- Each clip is 1 second long, sampled at 16 kHz

Three audio classes:

- **Stationary Background**
  (fan, AC, computer hum, quiet room)
- **Speech Dominant**
  (conversations, discussions, lectures)
- **Non-Speech Activity**
  (typing, footsteps, coughing, clapping)



Methodology - [12] Fused Disturbance Score

$$\textbf{Score} = 100 \times \text{clamp}(W_L L + W_S S + W_P P + W_I I,\ 0.0,\ 1.0)$$

Loudness Weight ($W_L = 0.45$)

- Contribution: 45% of total score.

Physical sound pressure (LAeq) is the strongest indicator of disturbance.

Persistence Weight ($W_P = 0.20$)

- Contribution: 20% of total score.

How continuously the sound is present over time.

Speech Weight ($W_S = 0.25$)

- Contribution: 25% of total score.

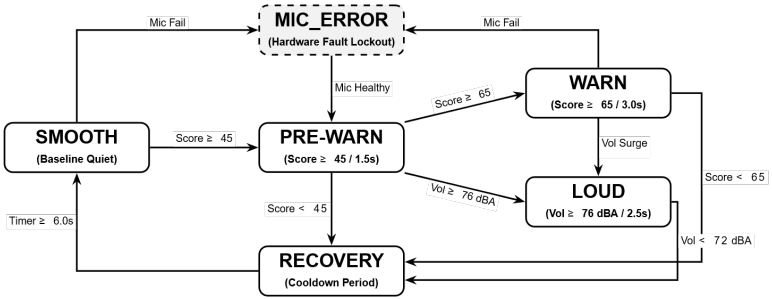
Using TinyML CNN probability to identify human speech.

Impulsiveness Weight ($W_I = 0.10$)

- Contribution: 10% of total score.

Represents short, non-speech impact events.

Methodology - [13] State Machine Architecture



State & Threshold Calibration :

- 4 states: SMOOTH → PRE-WARNING → WARNING → LOUD
- Thresholds are tested using measured audio and reference sound-level meter.
- ISO 22955:2021 / ANSI/ASA S12.60 used as acoustic-comfort references.
- Thresholds tuned for stable transitions and fewer false alarms.

Results - [1] Model Performance

Test Set Confusion Matrix Comparison (1,876 Samples)

Float32 Baseline (Before)

True \ Pred	BG	Speech	Non-Sp
Background	**323**	3	89
Speech	4	**492**	16
Non-Speech	68	12	**869**

Full INT8 Model (Deployed)

True \ Pred	BG	Speech	Non-Sp
Background	**324**	4	87
Speech	4	**493**	15
Non-Speech	69	12	**868**

Quantization Impact

Full INT8 post-training quantization preserves decision boundaries across all three classes with negligible impact on accuracy (**89.77%** $\rightarrow$ **89.82%**) while reducing model memory footprint to **75.6 KiB**.

Results - [2] Local Web Dashboard



Live state, score, class probabilities, calibrated level, baseline, and diagnostics.

Results - [3] Integrated Prototype



Integrated functions

- I2S capture
- DSP and INT8 CNN
- TFT state display
- RGB feedback
- Local dashboard

Validation - [1] Test Scenarios

ID	Scenario	Range	Target	Expected
VAL-01	Standard conversation	58–66 dBA	speech dominant	WARNING
VAL-02	Quiet study	$<$ 50 dBA	background	SMOOTH
VAL-03	AC / ceiling fan	52–56 dBA	background	SMOOTH/PRE-WARNING
VAL-04	Clap / object drop	$\geq$ 78 dB peak	DSP spike	SPIKE
VAL-05	Group shouting	$\geq$ 76 dBA	DSP loud	LOUD
VAL-06	Speech stops	$<$ 53 dBA	background	RECOVERY

Validation purpose

Verify that physical level, detected class, fused score, temporal logic, and final visual state remain consistent under representative acoustic conditions.

Validation - [2] Measured Data

Scenario	L_{Aeq}	Peak	Score	Final state
Conversation	73.4	90.8	68	LOUD
Quiet baseline	43.1	49.8	34	SMOOTH
Fan	53.8	59.8	49	PRE-WARNING
Handclap / spike	55.2	82.4	22	SPIKE
Speech stops	46.2	52.8	10	RECOVERY

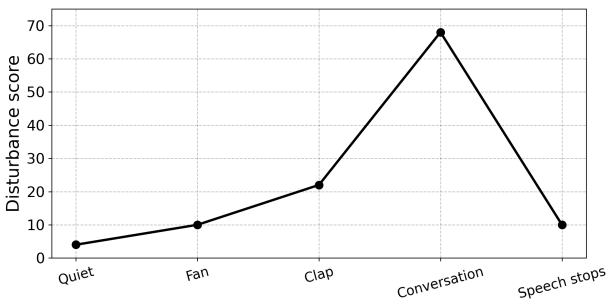
Observed class outputs

- Speech dominant
- Stationary background
- Non-speech activity

Calibration check

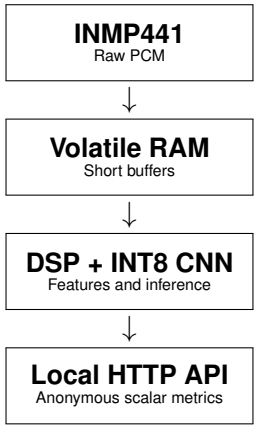
- Tested from 50 to 80 dBA
- Maximum absolute error: approximately 0.4 dB

Validation - [3] State Logic Matched Expectations



- Conversation produced WARNING.
- Quiet and fan remained SMOOTH.
- Clap produced the SPIKE overlay.
- Speech stopping triggered RECOVERY.
- Observed logic matched expected states.

Privacy-by-Design Boundary



Privacy guarantee

Audio is processed locally on the ESP32-S3. Raw buffers are overwritten and never written to flash, SD storage, or the network.

- No conversation recordings
- No raw-audio upload endpoint
- Only class probabilities, levels, states, and diagnostics are exposed

Future Enhancements

- Develop a dedicated PCB and compact enclosure with improved acoustic isolation.
- Add adaptive room calibration and automatic threshold adjustment.
- Expand the dataset with diverse Nepalese classroom and office recordings.
- Add more acoustic classes, secure MQTT telemetry, and multi-device monitoring.
- Develop a companion mobile application and secure over-the-air updates.
- Pursue standardized acoustic testing and Class 2 sound-level certification.

Conclusion

75.6 KiB	89.82%	46,867	100 ms
deployed INT8 model	test accuracy	model parameters	fusion update period

- Project runs real-time acoustic analysis locally on an ESP32-S3.
- Hybrid DSP and TinyML separate physical loudness from acoustic context.
- Temporal logic provides stable, non-judgmental visual feedback.
- The design avoids raw-audio retention and transmission by default.
- Prototype, dashboard, calibration, and scenario tests demonstrate feasibility.

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THANK YOU!

Questions and Discussion

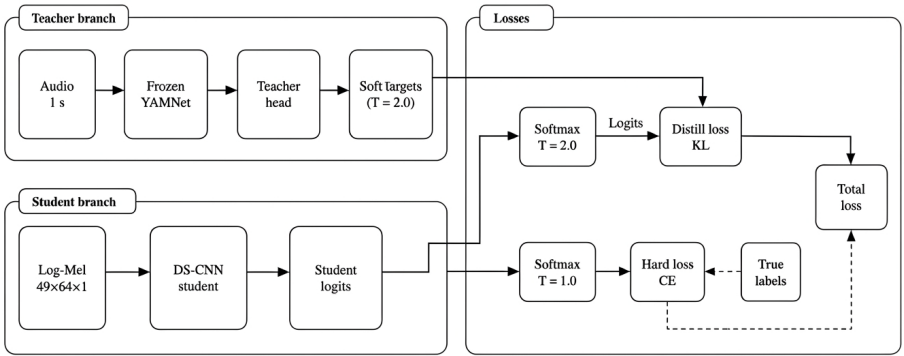
Privacy-Preserving Acoustic Intelligence at the Edge

APPENDIX

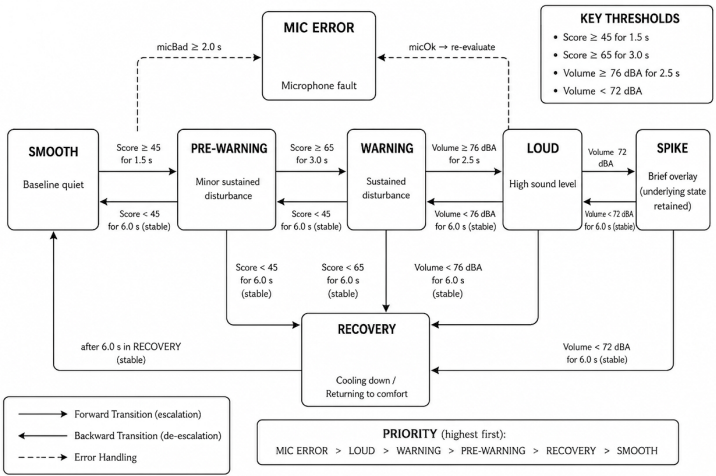
Technical Architecture and Implementation

Supplementary Material for Defense Q&A

Appendix – DS-CNN Architecture

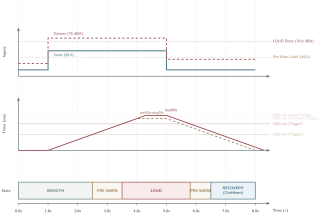


Appendix – Acoustic State Machine

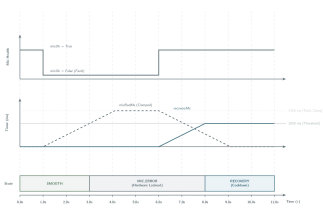


Appendix – State Transition Timing

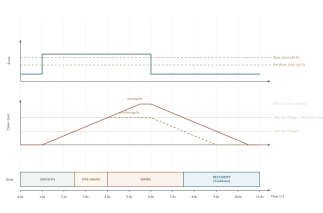
LOUD State Response



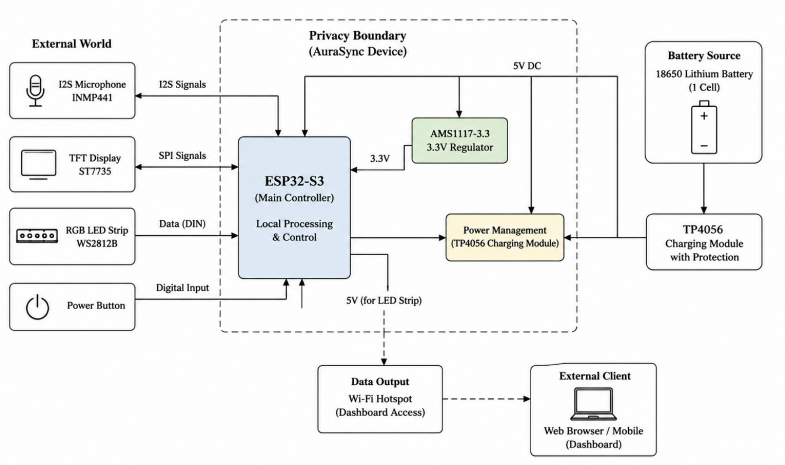
Mic Fault / Error Handling



Warning / Pre-Warning



Appendix – Privacy-Preserving Architecture



Appendix – Circuit Diagram

